

1.5 LARMOR FREQUENCY

The natural frequency of a precessing proton

A proton precessing in the main magnetic field (B_0) has a natural frequency called the Larmor frequency (f_0). When RF energy at this exact frequency is applied, the proton absorbs energy efficiently. This is the principle of **RESONANCE**.

THE CONCEPT

Every proton precesses at a frequency that depends only on the strength of the main magnetic field (B_0) and a constant called the gyromagnetic ratio (γ).

This frequency is called the Larmor frequency (f_0).

LARMOR FREQUENCY EQUATION

$$f_0 = \frac{\gamma}{2\pi} B_0$$

WHERE:

f_0 = Larmor frequency (Hz)

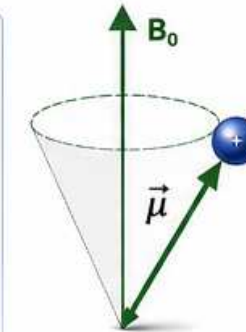
γ = Gyromagnetic ratio (rad/s/T)

B_0 = Main magnetic field (Tesla)

2π = Conversion from rad/s to Hz

For Hydrogen (^1H):

$$\frac{\gamma}{2\pi} = 42.58 \text{ MHz/T}$$



A proton precesses around the B_0 field axis at the Larmor frequency (f_0).

WHY LARMOR FREQUENCY MATTERS

RF excitation works only when the RF frequency (f_{RF}) matches the proton's Larmor frequency (f_0).

TOO LOW
($f_{RF} < f_0$)

RF is too slow.

Protons cannot keep up → little to no energy absorbed.

Inefficient excitation

MATCH
($f_{RF} = f_0$)

RF matches Larmor frequency.

Protons absorb energy efficiently → strong excitation.

Maximum energy absorbed

TOO HIGH
($f_{RF} > f_0$)

RF is too fast.

Protons cannot follow → little to no energy absorbed.

Inefficient excitation

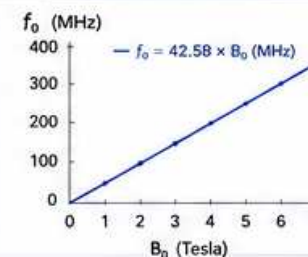
Exact match = **RESONANCE** = Efficient energy transfer

LARMOR FREQUENCY OF ^1H AT COMMON FIELD STRENGTHS

Magnetic Field (B_0)	Field Strength	Larmor Frequency (f_0)
0.1 Tesla	100 mT	4.26 MHz
0.2 Tesla	200 mT	8.52 MHz
0.5 Tesla	500 mT	21.29 MHz
1.0 Tesla	1.0 T	42.58 MHz
1.5 Tesla	1.5 T	63.87 MHz
3.0 Tesla	3.0 T	127.74 MHz
7.0 Tesla	7.0 T	298.06 MHz

KEY POINTS

- Larmor frequency is directly proportional to B_0 .
- Higher field strength → higher frequency.
- This is why 3T systems use higher RF frequency than 1.5T systems.



RESONANCE EXAMPLE (1.5 TESLA SYSTEM)

At 1.5 Tesla:



What happens?

- RF = 63.87 MHz** Protons absorb energy → excitation occurs.
- RF < 63.87 MHz** Protons don't absorb → weak or no signal.
- RF > 63.87 MHz** Protons don't absorb → weak or no signal.

MRI systems are carefully tuned so the RF frequency always matches the Larmor frequency at the selected field strength.

ENGINEERING INSIGHT

- The magnet determines B_0 .
- The RF system must be tuned to the correct Larmor frequency.
- If the frequency is off, you will not get the expected signal.



FIELD SERVICE TIP

- Verify correct field strength after magnet service or shim.
- Use the system's software tools to confirm transmit frequency.
- Even small frequency errors reduce SNR.

COMMON MISCONCEPTION

"More RF power can compensate for wrong frequency."

Reality: If the frequency is not correct, more power will not create resonance. It may only heat tissue.

REAL-WORLD ANALOGY

Think of a child on a swing. If you push at the right rhythm (matching the natural frequency), the swing goes higher. If you push too slow or too fast, the swing hardly moves.



KEY TAKEAWAY

The Larmor frequency is the natural frequency of a proton in the B_0 field. RF excitation is effective only when the RF frequency exactly matches the Larmor frequency. This is the foundation of MRI signal generation.

WHAT'S NEXT? >>

Next: Section 1.6 – RF Excitation & Flip Angle
We'll see how RF energy at the Larmor frequency tips the net magnetization away from the B_0 axis.